

Andrea Gilardi | Curriculum Vitae

– Milan, Italy

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Summary

I am a tenure-track assistant professor (RTT) in Statistics (S.S.D. STAT-01/A) at the [Department of Economics, Management and Statistics](#) of [University of Milano - Bicocca](#). My research focuses on the analysis of spatial and spatio-temporal data, with a particular interest on linear networks and environmental problems.

Current Position

University of Milano - Bicocca
Tenure Track Assistant Professor (RTT)

Milan (IT)
FEB. 25 - Today

Education

University of Milano - Bicocca
PhD in Statistics and Mathematical Finance

NOV.17–APR.21

University of Milano - Bicocca
Master Degree in Statistics (LM-82)
Final mark: 110/110 e lode (Magna cum laude)

SET.15–OTT.17

Università of Milano - Bicocca
Bachelor Degree in Statistics (L-41)
Final mark: 110/110 e lode (Magna cum laude)

SET.12–SET.15

Teaching

Lecturer (Docente)

- *Statistics and Data Analysis*, Politecnico di Milano, Bachelor Degree in Interaction Design, 28 hours, a.y. 2025-2026.
- *Sampling Methods*, University of Milano - Bicocca, Master Degree in Statistics and Economics, 42 hours, aa.yy. 2025-2026; 2026-2027.
- *Statistics I*, University of Milano - Bicocca, Bachelor degree in Marketing, Comunicazione Aziendale e Mercati Globali, 40 hours, a.y. 2024-2025.
- *Lab - Data Science for Industrial Engineering*, Politecnico di Milano, Master degree in Mechanical Engineering, 45 hours, a.y. 2024-2025, Co-Lecturer.
- *Statistical Learning for Healthcare Data*, Politecnico di Milano, Master degree in Mathematical Engineering, 30 hours, a.y. 2024-2025.
- *Basics of Statistics and Biomedical Signals*, Politecnico di Milano, Bachelor degree in Biomedical Engineering, 30 hours, a.y. 2023-2024.
- *R for Data Science*, University of Milano - Bicocca, PhD in Economics, Management and Statistics, Curriculum Statistics, 12 hours per year, aa.yy. 2022-2023; 2023-2024.
- *Livellamento al Software R* (Introduction to the R Software), University of Milano - Bicocca, Master degree in Statistics and Economics, 14 hours per year, aa.yy. 2022-2023; 2023-2024; 2024-2025.

Teaching Assistant (Esercitatore)

- *Basics of Statistics and Biomedical Signals*, Politecnico di Milano, Bachelor degree in Biomedical Engineering, 15 hours, a.y. 2023-2024.
- *Statistica II*, University of Milano - Bicocca, Bachelor degree in Statistical and Economic Sciences, 20 hours per year, aa.yy. 2017-2018; from a.y. 2022-2023 to 2025-2026.

Tutor

- *Statistica II*, University of Milano - Bicocca, Bachelor degree in Statistical and Economic Sciences, 40 hours per year, aa. yy. 2018-2019; 2019-2020; 2020-2021; 2021-2022.
- *Lab Data Challenges*, University of Milano - Bicocca, Master degree in Statistics and Economics, 20 hours, a.y. 2020-2021.
- *R for Road Safety*, RAC Foundation, 12 hours, Oct. 2019, [Program](#) and [Materials](#).
- *Introduction to R*, University of Milano - Bicocca, Bbetween 2019 MULTIMEDIA, 20 hours, a.y. 2018-2019, [Materials](#).
- *Introduction to tidyverse and plumber*, University of Turin - Master in Data Science, 4 hours, Oct. 2018.
- *Statistica Economica M* (Economic Statistics), University of Milano - Bicocca, Master degree in Statistics and Economics, 20 hours, a.y. 2017-2018.
- *Informatica* (Informatics), University of Milano - Bicocca, Bachelor degree in Statistical and Economic Sciences, 20 hours, a.y. 2016-2017.

Working Experience

Politecnico di Milano

Assistant Professor (RTDA)

Milan (IT)

JUN. 23 - FEB. 25

I worked on a project named [GRINS: Growing Resilient, INclusive and Sustainable](#) which is part of the Next Generation EU program (National Recovery and Resilience Plan).

POLIMI Graduate School of Management

Instructor

Milan (IT)

NOV. 23 - DEC.23

I held a 16 hours course entitled "Analytics - Tools di Analisi" for approximately 30 employees of [ACEA](#). We focused on descriptive and exploratory statistics, supervised and unsupervised clustering techniques, and principal components analysis with practical examples in JASP.

University of Milano - Bicocca

Research fellow (Assegnista di Ricerca)

Milan (IT)

JAN. 21 - JUN. 23

I worked on a project entitled "Spatio-temporal models for spatial network data" under the supervision of Prof. [Riccardo Borgoni](#).

Catholic University of the Sacred Heart

Tutor

Milan (IT)

OCT.20-NOV.20

I worked on a project named *Black Box and Big Data per la stima e la previsione dei sinistri* under the supervision of Prof. [Diego Zappa](#).

University of Milano - Bicocca

Tutor

Milan (IT)

JUN.20-OCT.20 & JUN.21 - OCT.21 & JUN.22 - OCT.22 & JUN.23-OCT.23

I prepared and organized the admission tests for the master degree in Statistics and Economics, Department of Economics, Management and Statistics.

GR & Partner S.A.S

Consultant

Milan (IT)

APR.18-OCT.18

I developed a User Interface using R and Shiny.

PhD Thesis

Title: *Statistical Models and Data Structures for Spatial Data on Road Networks*

Supervisor: Prof. Riccardo Borgoni and Prof. Jorge Mateu

Summary: The thesis introduces data structures, vital pre-processing steps, and statistical models to analyse spatial data lying on road networks using point-pattern and network-lattice approaches.

Master Degree Thesis

Title: *Previsione spaziotemporale per gli interventi in emergenza delle ambulanze nel comune di Milano*

Supervisor: Prof. Riccardo Borgoni

Summary: We developed a spatiotemporal model to predict the distribution of the emergency medical interventions in Milan.

Visiting Periods

- I spent seven months in Castellón de la Plana (ES), from February 2020 to June 2020 and May 2021 to July 2021, working with Prof. [Jorge Mateu](#) on statistical models for events on road networks.
- I stayed in Leeds (UK) for three months, from September 2019 to November 2019, working with Prof. [Robin Lovelace](#) on R packages and classes for road networks. [Final presentation](#).

Publications

Articles in refereed journals.....

1. **Gilardi, A.**, Borgoni, R., Mateu, L. (2026+) Spatial calibration of mobile sensor traffic data on road networks. *Quality and Reliability Engineering International* Accepted.
2. Scotti, F., Bonaccorsi, G., Buongiorno, A., **Gilardi, A.**, Intini, M., Urbano, V. M., Bergantino, A. S., (2026) Local environmental gains from green Cohesion Policy: a study of air pollution reduction in Italian municipalities. *Regional Studies* [[10.1080/00343404.2026.2723297](#)].
3. Carlesso, L. M., Cappozzo, A., **Gilardi, A.**, Zuccolotto, P. (2026). A mixed-effects spatial point process framework for modeling player shooting behavior in basketball. *Journal of Quantitative Analysis in Sports* [[10.1515/jqas-2025-0163](#)].
4. Borgoni, R., **Gilardi, A.**, Michelangeli, A., Tkach, K. (2025). An Inverted-U, an Elephant or Neither? Exploring the Link Between Migrant Integration Policies and Pro-Immigrant Sentiment. *POPULATION SPACE & PLACE* [[10.1002/psp.70053](#)].
5. D'Angelo, N., Albano, A., **Gilardi, A.**, Adelfio, G. (2025) Non-separable spatio-temporal Poisson point process models for fire occurrences. *Environmental Ecological Statistics* [[10.1007/s10651-025-00645-x](#)].
6. **Gilardi, A.**, Borgoni, R., Mateu, J. (2024) A non-separable first-order spatio-temporal intensity for events on linear networks: an application to ambulance interventions. (2024) *Annals of Applied Statistics* [[10.1214/23-AOAS1800](#)].
7. **Gilardi, A.**, Borgoni, R., Presicce, L., Mateu, J. (2023) Measurement Error Models for Spatial Network Lattice Data: Analysis of car crashes in Leeds. *Journal of the Royal Statistical Society: Series A (Statistics in Society)* [[10.1093/jrssa/qnad057](#)].
8. **Gilardi, A.**, Mateu, J., Borgoni, R., Lovelace, R. (2022) Multivariate hierarchical analysis of car crashes data considering a spatial network lattice. *Journal of the Royal Statistical Society: Series A (Statistics in Society)* [[10.1111/rssa.12823](#)];
9. Borgoni, R., **Gilardi, A.**, Zappa, D. (2020). Assessing the Risk of Car Crashes in Road Networks. *Social Indicators Research* [[10.1007/s11205-020-02295-x](#)];

Conference papers and proceedings.....

1. **Gilardi, A.** (2026). Analysing the Relationship Between Traffic Flows, Road Infrastructure, and Car Crashes Data: An Approach Based on Spatiotemporal Point Patterns on Linear Networks. In: Chelli, F.M., Crocetta, C., Ingrassia, S., Recchioni, M.C. (eds) *Statistical Learning, Sustainability and Impact Evaluation*. SEAS IN 2023. Springer Proceedings in Mathematics & Statistics, vol 523. Springer, Cham. https://doi.org/10.1007/978-3-032-10630-8_16
2. **Gilardi, A.**, De Sanctis, M., Milan, G., Ieva, F., Sangalli, L., Secchi, P. (2025). Three Spatial Methods for Assessing PM₁₀ Concentration in the Lombardy Region. In *Statistics for Innovation I SIS 2025, Short Papers, Plenary, Specialized, and Solicited Sessions* (pp.338-343). Springer. https://dx.doi.org/10.1007/978-3-031-96736-8_56.
3. **Gilardi, A.**, Ieva, F., Statistical methods for evaluating the environmental sustainability. (2024) Book of short papers SIS (2024). *Methodological and Applied Statistics and Demography IV*. Springer. Electronic ISSN: 3059-2143. Print ISBN: 978-3-031-64446-7 https://doi.org/10.1007/978-3-031-64447-4_18;
4. D'Angelo, N., Albano, A., **Gilardi, A.**, Adelfio, G., Spatio-temporal modelling of fires in Sicily. (2024) Book of short papers SIS 2024. Springer. Electronic ISSN: 3059-2143. https://doi.org/10.1007/978-3-031-64350-7_60;
5. **Gilardi, A.**, Borgoni, R., The impact of traffic flow and road signs on road accidents: an approach based on spatiotemporal point pattern analysis on linear networks. In: *Book of the Short Papers - SIS 2023* (pp. 702-707), Pearson, ISBN: 9788891935618AAVV, Ancona (IT). [Link](#);
6. **Gilardi, A.**, Borgoni, R., Mateu, J. (2022). Geographically weighted regression for spatial network data: an

- application to traffic volumes estimation. In 51th Scientific Meeting of the Italian Statistical Society (pp. 1504 - 1509). Torino:Pearson Italia. ISBN: 9788891932310, Caserta (IT). [Link](#).
7. **Gilardi, A.**, Borgoni, R., Mateu, J. (2022). Spatial statistical calibration on linear networks: an application to the analysis of traffic volumes. In Proceedings of the 10th International Workshop on Spatio-Temporal Modelling, Lleida (Spain), 1-3 June 2022. ISBN 978-84-9144-364-3. http://doi.org/10.21001/METMA_X.
 8. Borgoni, R., **Gilardi, A.**, Zappa, D. (2022). Optimal Subgrids from Spatial Monitoring Networks. In: (a cura di): Lombardo R; Camminatiello I; Simonacci V, IES 2022 Innovation & Society 5.0: Statistical and Economic Methodologies for Quality Assessment - Book of Short Papers. p. 148-152, Sesto San Giovanni:Professional Knowledge Empowerment, ISBN: 978-88-94593-35-8, Capua, 2022. [Link](#).
 9. **Gilardi, A.**, Borgoni, R., Presicce, L., Mateu, J. (2021) Measurement error models on spatial network lattices: car crashes in Leeds. In 13th Scientific Meeting of the Classification and Data Analysis Group, Firenze, September 9-11, 2021. <https://doi.org/10.36253/978-88-5518-340-6>;
 10. **Gilardi, A.**, Borgoni, R., Mateu, J. (2021). A spatio-temporal model for events on road networks: an application to ambulance interventions in Milan. In 50th Scientific Meeting of the Italian Statistical Society (pp. 702 - 707). Torino:Pearson Italia. ISBN: 9788891927361, Pisa - Virtual. [Link](#);
 11. Borgoni, R., **Gilardi, A.**, Zappa, D. (2020). Spatial modelling of driver crash risk using georeferenced data. In NESPUTT 2019 - New Economic & Statistical Perspectives on Urban & Territorial Themes. (NESPUTT 2019). E-book. vol. libreriauniversitaria, ISBN: 9788833592091, University of Milan - Bicocca, 2019. [Link](#);
 12. **Gilardi, A.**, Borgoni, R., Zappa, D. (2020). Assessing the Risk of Car Accidents in Road Networks. In NESPUTT 2019 - New Economic & Statistical Perspectives on Urban & Territorial Themes. (NESPUTT 2019). E-book. vol. libreriauniversitaria, ISBN: 9788833592091, University of Milan - Bicocca, 2019. [Link](#);
 13. **Gilardi, A.**, Borgoni, R., Zappa, D. (2019). Spatial Logistic Regression for Events Lying on a Network: Car Crashes in Milan. In: Book of Short Papers SIS2019. (pp. 1165-1170), Pearson, ISBN: 9788891915108, Milano, 2019. [Link](#);
 14. **Gilardi, A.**, Borgoni, R., Pagliosa, A., Bonora, R. (2018). Spatiotemporal Prevision for Emergency Medical System Events in Milan. In: Book of short Papers SIS 2018 (pp. 1697-1702). ISBN: 9788891910233, Palermo, Italia, 2018. [Link](#).

Summer Schools and Workshops

- *2nd Meeting on Territorial Sustainability, Spoke 7 (GRINS)*, University of Bari (IT), February 13-14, 2025
- *GRINS Workshop: Geostatistical harmonisation of large environmental and climate datasets for the AMELIA platform: inter-spoke coordination and first results*, University of Bergamo (IT), March 22nd, 2024.
- *R Project Sprint 2023*, University of Warwick (UK), August 30 - September 1, 2023. [Webpage](#).
- *Contributing to R*, Online workshop held during useR! 2021.
- *Spatial Statistics for huge datasets and best practices*, Online workshop held during useR! 2021.
- *Tidyverse developer day*, Tolouse (FR), July 2019. [Announcement](#).
- *Robust Statistics : Foundations and Recent Developments*, University of Milano - Bicocca (IT), March 2019.
- *CLADAG, Summer School on Clustering and Classification*, University of Catania (IT), May 2018.

Conferences

Invited.....

- *65th ISI World Statistics Congress 2025*, October 5-9 2025, The Hague (NL)
- *Spatial Data Science across Languages (2025)*, September 17 - 19 2025, Paris Lodron University Salzburg, Department of Geoinformatics, Salzburg (AT)
- *Workshop on Nonparametric Statistics*, May 15-16 2025, Luiss University, Rome (IT)
- *Stat Data Camp*, September 23-27 2024 - University of Padova (IT). This was a workshop organised by the University of Padova in Bressanone (IT) where I presented my work on the R package `sfnetworks`.
- *Statalk*, May 30-31 2024, Department of Statistics, Computer Science, Applications G. Parenti, University of Florence (IT)
- *2023 IMS International Conference on Statistics and Data Science (ICSDS)*, December 18-21 2023, Lisbon (PT)

- *15th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics)*, December 17-19 2022, King's College London, London (UK).
- *Summer Institute in Computational Social Science (SICSS-Covenant)*, June 22-29 2022, Covenant University, Ogun State (NG).
- *OpenGeoHub Summer School 2021*, September 1-3 2021. Online Event.

Contributed

- *Spatial Statistics 2025: At the Dawn of AI*, July 15-18 2025, Noordwijk (NH)
- *16th Workshop on Spatial Statistics and Image Analysis in Biology*, June 2-5 2025, Smogen (SE)
- *METMA XI, 11th International Workshop on Spatio-Temporal Modelling*, July 23-25 2024, Lancaster (UK)
- *useR! 2024 Meeting*, July 8-11, Salzburg (AU)
- *52th meeting of the Italian Statistical Society*, June 17-20 2024, Bari (IT)
- *Free and Open Source Software for Geospatial (FOSS4G)*, August 22-28 2022, Florence (IT).
- *51th meeting of the Italian Statistical Society*, June 22-24 2022, Caserta (IT).
- *METMA X, 10th International Workshop on Spatio-Temporal Modelling*, June 1-3 2022, Lleida (ES)
- *CLADAG 2021, 13th Scientific Meeting Classification and Data Analysis Group*, September 9-11 2021, Online
- *UseR! 2021 Meeting*, July 5-9 2021, Online
- 50th meeting of the Italian Statistical Society, 21-25 June 2021, Online.
- *OpenGeoHub 2020*, August 16-23 2020, Wageningen (NH).
- *Nesputt 2019: New Economic & Statistical Perspective*, November 21-22 2019, Milan (IT).
- *Meeting of the Italian Statistical Society*, June 18-21 2019, Milan (IT).
- *49th Meeting of the Italian Statistical Society*, June 20-22 2018, Palermo (IT).

Attended

- *2024 IMS International Conference on Statistics and Data Science (ICSIDS)*, December 16-19 2024, Nice (FR)
- *StataTalk*, September 15th 2023, Università di Roma - Sapienza (IT).
- *GRASPA 2023*, July 10-11 2023, Palermo (IT).
- *Workshop of Young Environmental Statistician (YES)*, February 3rd, 2023, University of Bologna (IT).
- *e-Rum 2020: European R User Meeting*, June 17-20 2020, Online event.
- *UseR! 2019 Meeting*, July 9-12 2019, Toulouse (FR).
- *StataTalk*, March 29th, 2019, University of Bologna (IT).
- *StataTalk*, November 17th, 2017, University of Padova (IT).

Organized Conferences

- I was a member of the Organizing Committee for the [SDS 2025 Conference](#).
- I organised session for the CFE-CMStatistics 2024 conference which was held at King's College London (UK) from December, 14th to December, 16th 2024.
- I was a member of the Organizing Committee of the [Mathematics for planet Earth \(M4E\)](#) Workshop. This workshop was held on November 11-12, 2024 at Politecnico di Milano (IT)
- I organized an online Webinar and Hackathon related to the R package [sfnetworks](#) as official satellite event of the e-Rum 2020 Conference. [Materials](#), [slides](#) and [recordings](#).

Grants

- [Tidy Spatial Networks in R](#) funded by the [R Consortium](#). 9000\$. Joint project with [Lucas van der Meer](#), [Lorena Abad](#), and [Robin Lovelace](#).

Awards

- [IMS New Researcher Travel Award](#) to attend the ICSIDS 2024 Conference (750 USD)

- Travel award to attend the [R Project Sprint 2023](#) event. 650£.
- Scholarship for the OpenGeoHub 2020 - Summer School. Project: *plotKML functionality to support sf and stars objects*. 800€. [Final presentation](#).
- [SIS](#) award for the best poster at the SIS-2019 conference.
- [Young-SIS](#) award for the best poster at the SIS-2018 conference.

Community activities

Referee service

AStA Advances in Statistical Analysis (2); Journal of Transport Geography (1); PLOS ONE (2) Scientific Reports (3); Spatial Statistics (4); Statistical Methods & Applications (1); Stochastic Environmental Research and Risk Assessment (1); Technometrics (1).

Affiliations

International Statistical Institute (ISI); rOpenSci; Società Italiana di Statistica (SIS); Sezione Giovani della Società Italiana di Statistica (Y-SIS).

Projects and activities with public engagement

- Member of the group working on Progetto di Terza Missione entitled *Analisi della dislocazione spaziale e spaziotemporale degli interventi extraospedalieri e definizione di criteri per monitorare e dimensionare la flotta dei mezzi di soccorso*, University of Milano - Bicocca, 2023.
- *Corso di Formazione Tavolo territoriale / Elementi di geo-statistica e microsimulazione spaziale*. This is a workshop organised by [PoliS](#) and [Istat](#) that was held online and in presence at the end of June 2024. I presented a 90 minutes talk which was entitled *Incidentalità stradale in Lombardia*.
- Member of the national project named GRINS Growing Resilient, INclusive and Sustainable - Codice proposta PE00000018, CUP D43C22003110001, which is part of the Next Generation EU (National Recovery and Resilience Plan) program. In particular, we focused on analysing environmental data at the municipal level, with the final goal of developing an accurate and easy-to-communicate air quality index. Politecnico di Milano, June 2023 to February 2025.

Software

I am a proficient R user, with a particular interest in the `r-spatial` world and tidyverse environment. I am the author of two R packages, namely [osmextract](#) (reviewed by rOpenSci foundation and now on [CRAN](#)) and [sfnetworks](#) (also on [CRAN](#)). The first package focuses on the management and downloading of Open Street Map data extracts obtained from several providers (like [Geofabrik](#)), while the second package provides a tidy approach to spatial network analysis.

I contributed to several R packages like `ggplot2`, `recipes`, `sf`, `stplanr`, `dodgr`, and `osmdata`. I attended the [Tidyverse Developer Day](#) in Tolouse during the UseR! 2019 conference and I attended the [R Project Sprint 2023](#) event at the University of Warwick (UK).

I can write scientific documents and presentations using $\text{\LaTeX}$ and I have a basic knowledge of Rust, GDAL, POSTGIS, JASP, SAS, Python, and STATA.

Badges

SAS: [SAS Certified Base Programmer for SAS 9](#)

SAS: [SAS Certified Predictive Modeler Using SAS Enterprise Miner 7](#)

Additional Information

Associazioni: Volunteer at Volontari del Soccorso di Calolziocorte from July 2016.

Notes

Autorizzo la pubblicazione ai sensi del D.Lgs. n. 33/2013 "Riordino della disciplina riguardante gli obblighi di pubblicità, trasparenza e diffusione di informazioni da parte delle pubbliche amministrazioni" e acconsento all'utilizzo delle informazioni ivi contenute ai sensi del D.L. n. 196/2003 "Codice in materia di protezione dei dati personali".

Le dichiarazioni rese nel presente curriculum sono da ritenersi rilasciate ai sensi degli artt. 46 e 47 del D.P.R. 445/2000.